

Graphs of circles



1 Every point on the circle is exactly 5 units away from the point $(0,0)$.

2

a Inside

b On

c Outside

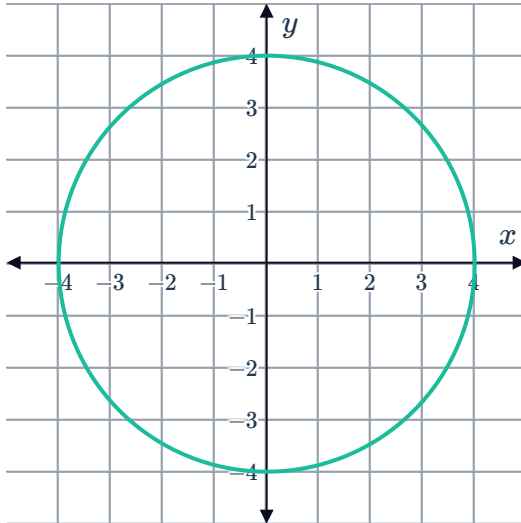
d On

3

a

i 4 units

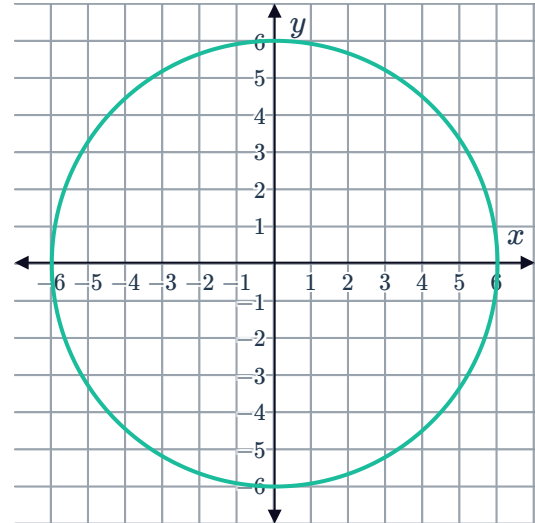
ii



b

i 6 units

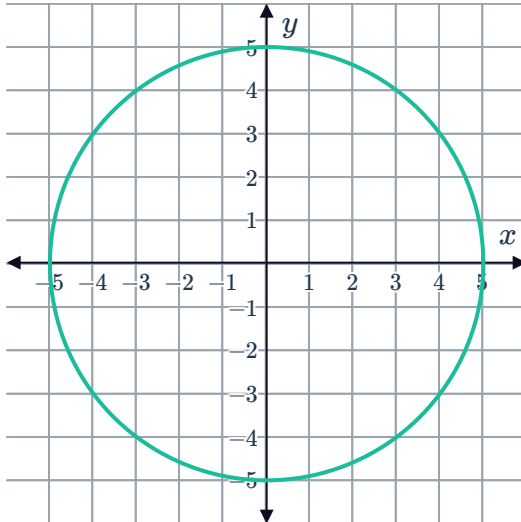
ii



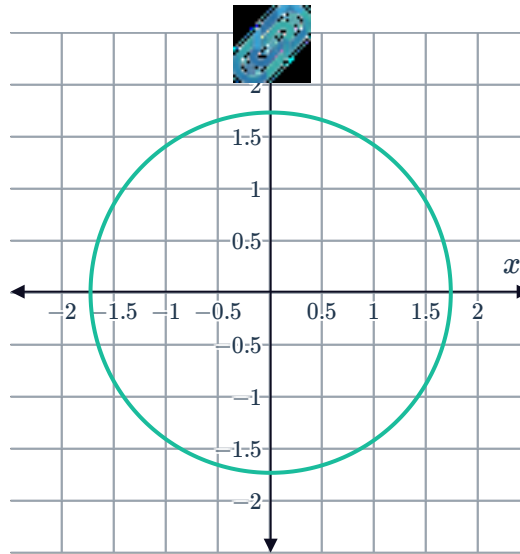
c

i 5 units

ii



4



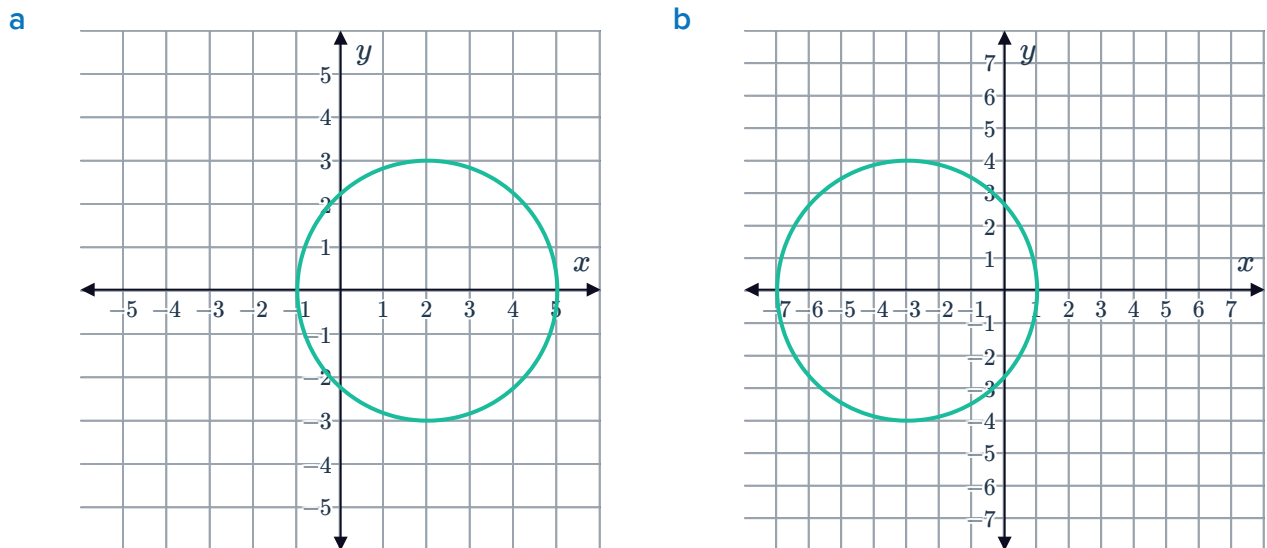
5

- a i $\left(-3.1, \frac{3}{2}\right)$ ii $\frac{3}{4}$ units
- b i $(6, 6)$ ii $2\sqrt{3}$ units
- c i $(0.7, -4.4)$ ii $\sqrt{3}$ units
- d i $(8, -1)$ ii $\sqrt{13}$ units

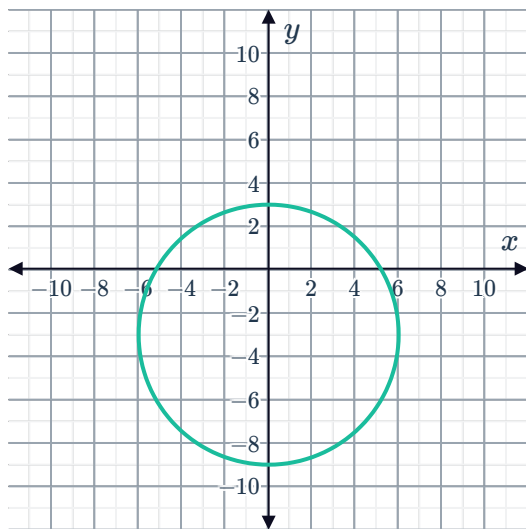
6

- a i $(0, 0)$ ii 2 units
- b i $(3, 2)$ ii 3 units
- c i $(1, -2)$ ii 8 units
- d i $(-3, -1)$ ii 5 units

7



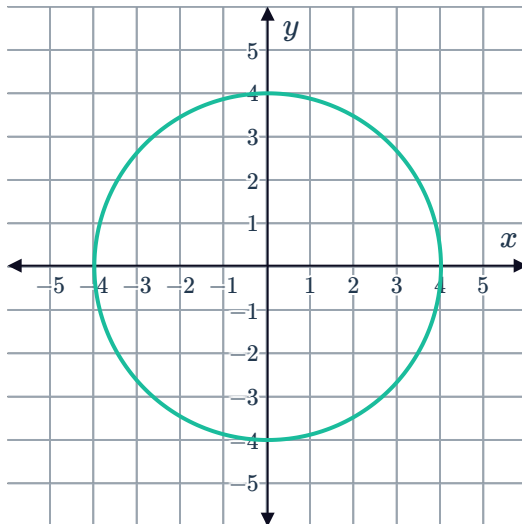
c



8

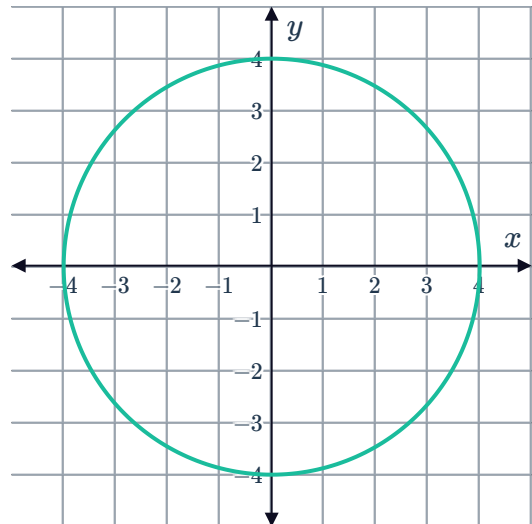
a

- i (0,0)
- ii 4 units
- iii



b

- i (0,0)
- ii 4 units
- iii

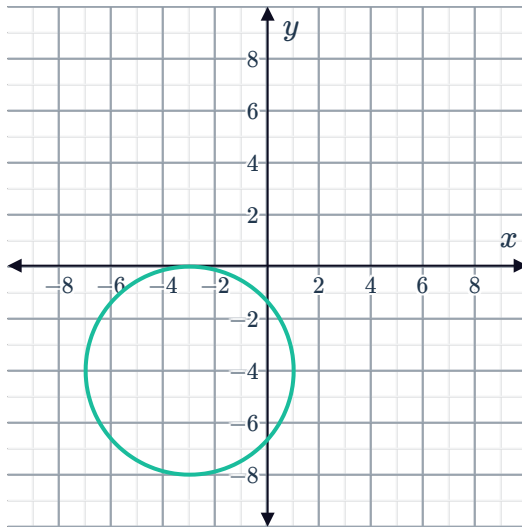


c

i $(-3, -4)$

ii 4 units

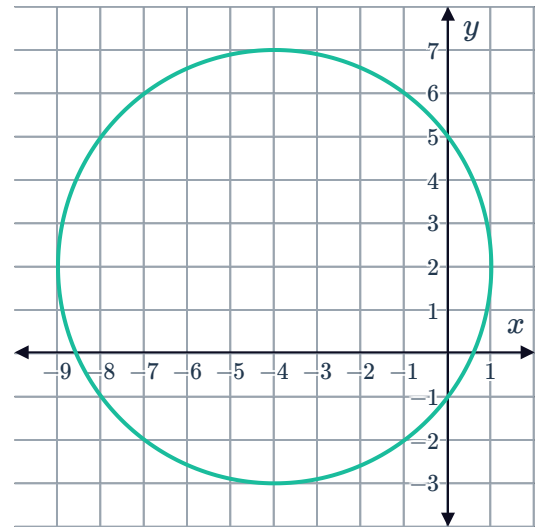
iii



i $(-4, 2)$

ii 5 units

iii

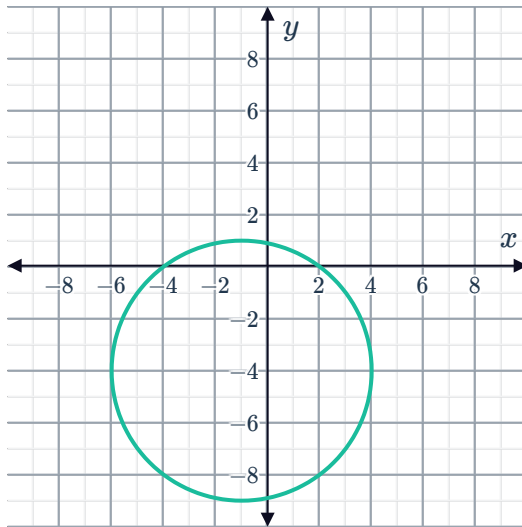


e

i $(-1, -4)$

ii 5 units

iii

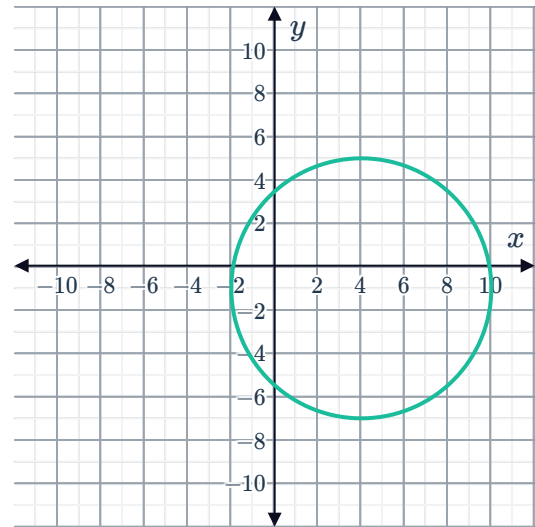


f

i $(4, -1)$

ii 6 units

iii

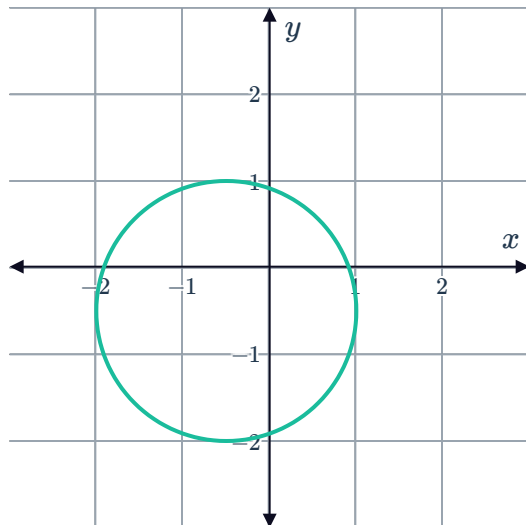


g

i $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

ii $\frac{3}{2}$ units

iii



9

a 41 units

b No

10 $(7, -6)$

11

a 5 units

b 5 units

c $(5, -1)$

d Yes

12

a $(6, 5)$

b 2 units

Equations of circles

13

a No

b No

c No

d Yes

e Yes

f Yes

14

a $5\sqrt{5}$ units

b $x^2 + y^2 = 125$

15

a $x^2 + y^2 = 36$



b $36\pi \text{ units}^2$

16 B

17

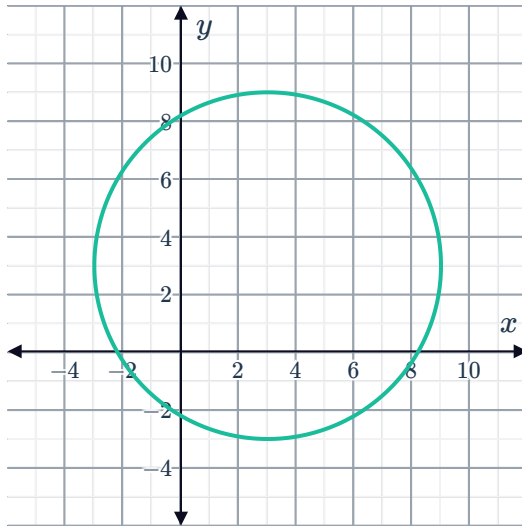
a $(1, 3)$

b 6 units

c $(x - 1)^2 + (y - 3)^2 = 36$

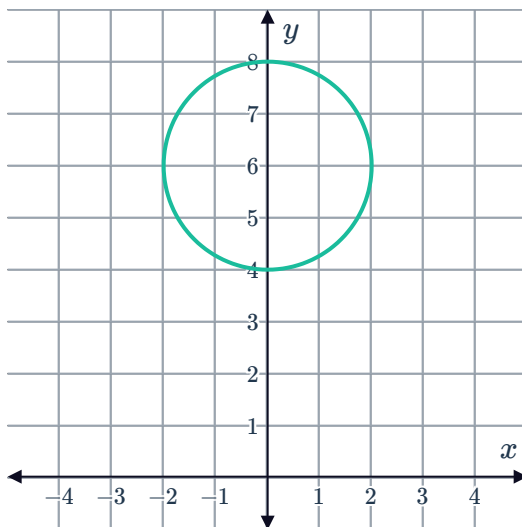
18

a i



ii $(x - 3)^2 + (y - 3)^2 = 36$

b i

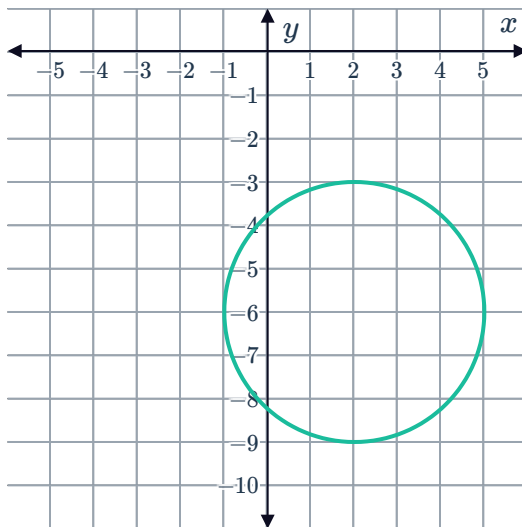


ii $x^2 + (y - 6)^2 = 4$

c
i

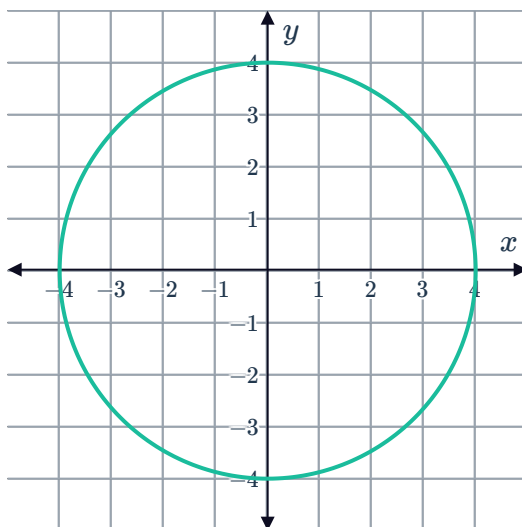


ii $x^2 - 4x + y^2 + 12y = -31$



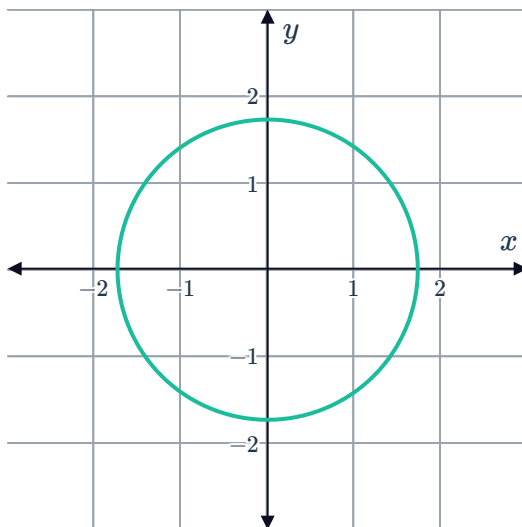
d
i

ii $x^2 + y^2 = 16$



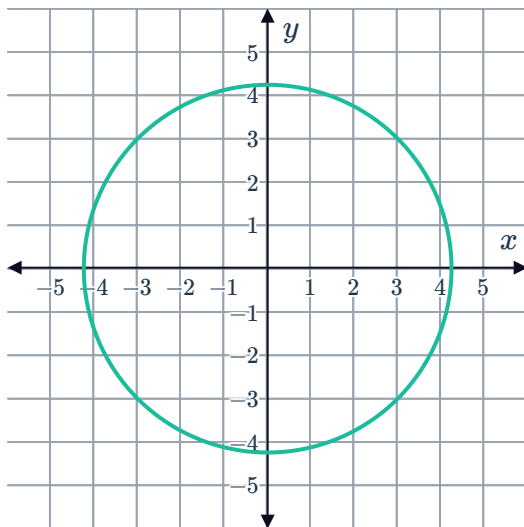
e
i

ii $x^2 + y^2 = 3$



f

i



ii $x^2 + y^2 = 18$

19

a $\sqrt{74}$ units

b $(x + 2)^2 + (y - 3)^2 = 74$

20

a

i $(x - 4)^2 + y^2 = 36$

ii $(4, 0)$

iii 6 units

b

i $(x + 2)^2 + (y + 3)^2 = 16$

ii $(-2, -3)$

iii 4 units

c

i $(x - 1)^2 + (y + 4)^2 = 36$

ii $(1, -4)$

iii 6 units

d

i $(x + 4)^2 + (y - 3)^2 = 1$

ii $(-4, 3)$

iii 1 units

21

a $\left(x - \frac{7}{2}\right)^2 + \left(y + \frac{5}{2}\right)^2 = 5$

b $(3.5, -2.5)$

c $\sqrt{5}$ units

d Inside

22

a

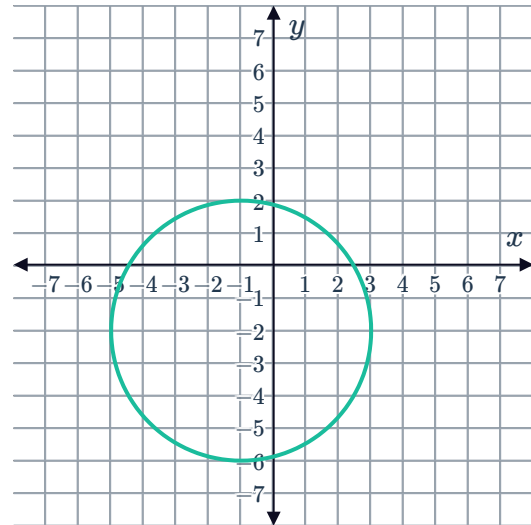
i $(x + 1)^2 + (y + 2)^2 = 16$

iii 4 units



ii $(-1, -2)$

iv



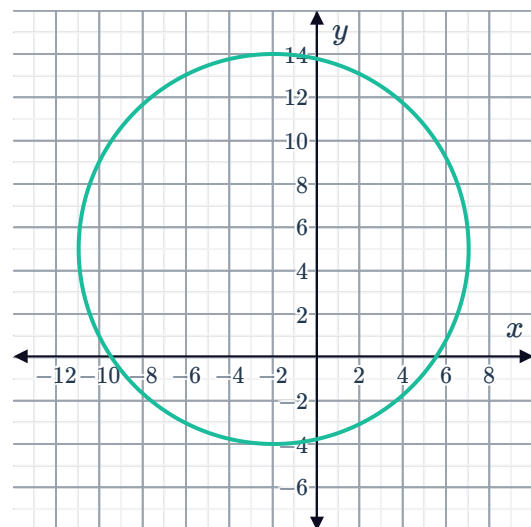
b

i $(x + 2)^2 + (y - 5)^2 = 81$

iii 9

ii $(-2, 5)$

iv



c

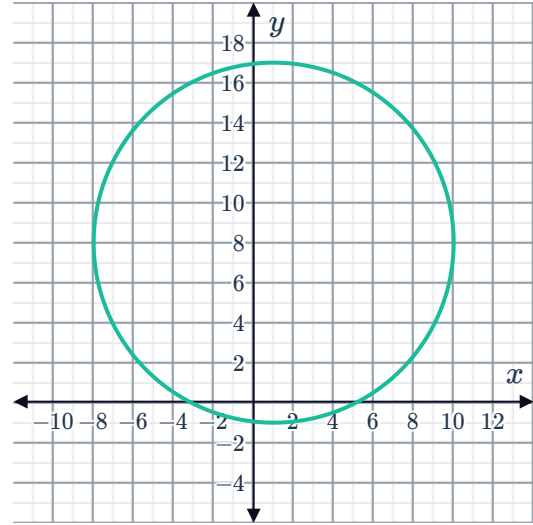
i $(x - 1)^2 + (y - 8)^2 = 81$

iii 9



ii (1, 8)

iv



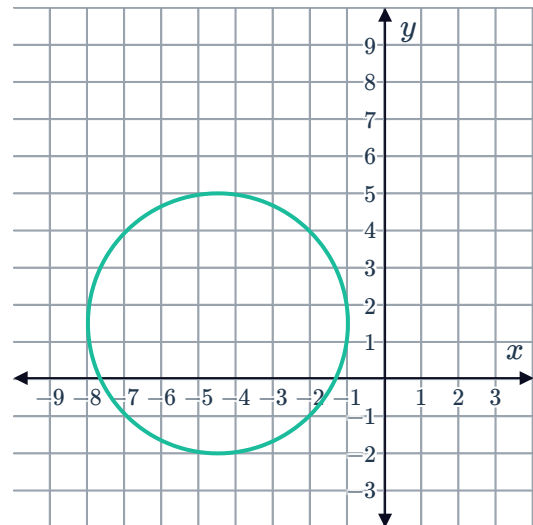
d

i $\left(x + \frac{9}{2}\right)^2 + \left(y - \frac{3}{2}\right)^2 = \frac{49}{4}$

iii $\frac{7}{2}$ units

ii $\left(-\frac{9}{2}, \frac{3}{2}\right)$

iv



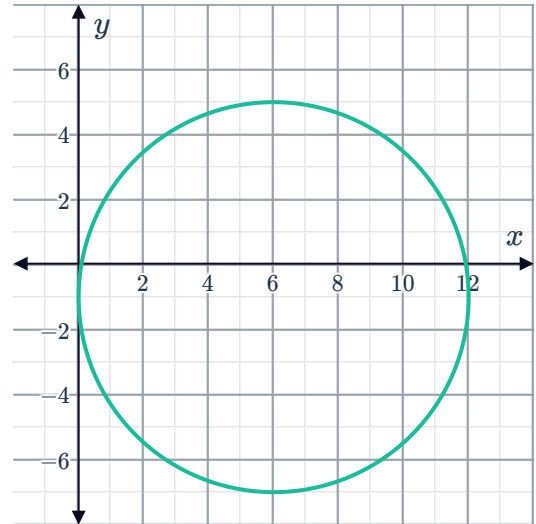
23

a $(x - 6)^2 + (y + 1)^2 = 36$

c 6 units

b $(6, -1)$

d



24

a $(-10, -9)$

b $\sqrt{61}$ units

c $(x + 10)^2 + (y + 9)^2 = 61$

Translations of circles

25 The graph of $(x + 3)^2 + y^2 = 4$ has been horizontally translated from the graph of $x^2 + y^2 = 4$ by 3 units to the left.

26

a $x^2 + (y + 2)^2 = 25$

b $x^2 + (y - 3)^2 = 7^2$

c $(x - 4)^2 + y^2 = 9$

d $(x + 5)^2 + y^2 = 4$

Domain and range of circles

27

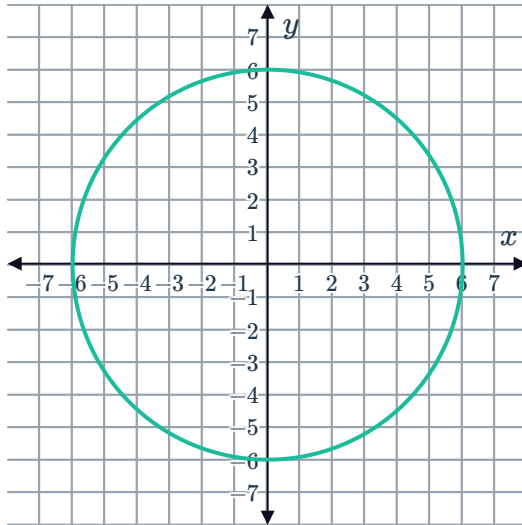
a i $[-6, 6]$ ii $[-6, 6]$

b i $[3, 9]$ ii $[-4, 2]$

28

29

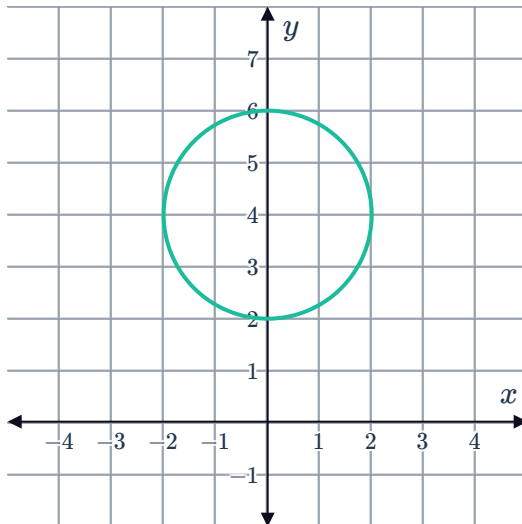
a
i



ii $[-6, 6]$

iii $[-6, 6]$

b
i



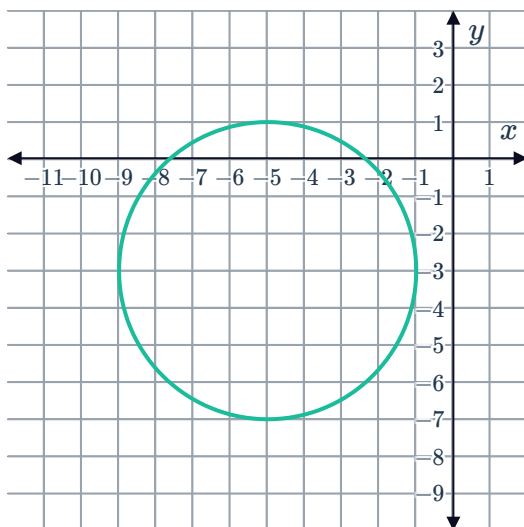
ii $[-2, 2]$

iii $[2, 6]$

c
i

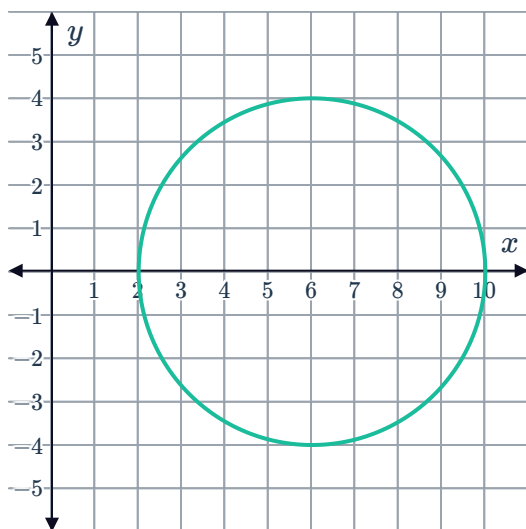


ii $[-9, -1]$



iii $[-7, 1]$

30
a



b $(x - 6)^2 + y^2 = 16$

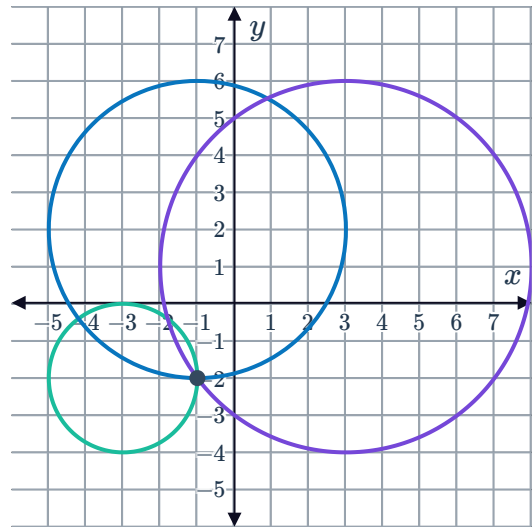
Applications

31
a 50 km

b No

32

- a As the point of intersection of the three circles centered at the receiving stations with radii as the distances, the epicenter is at $(-1, -2)$.



- b The point $(8, -7)$ is outside the circle with center $(-1, -2)$ with radius 10, so it would not feel the earthquake.

33

- a Domain: $[5, 19]$, Range: $[1, 15]$
 b Domain: $[14, 26]$, Range: $[2, 14]$
 c 5 cm
 d 85.19 cm^2
 e Circle one: $(x + 1)^2 + (y - 6.085)^2 = 49$, Circle two: $(x - 7)^2 + (y - 6.085)^2 = 36$

34 $x^2 + (y - 17)^2 = 144$

35 $(x + 5.3)^2 + (y + 5.4)^2 = 900$